Can we learn more when we cannot collect more data?

Counterfactual Residual Data Augmentation for Regression

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Proposal stage · no replication results yet

TODO: Group name / number · TODO: presentation date

Small, noisy tables are hard to learn from

01 Few samples

02 Costly to collect more

03 Noisy observations

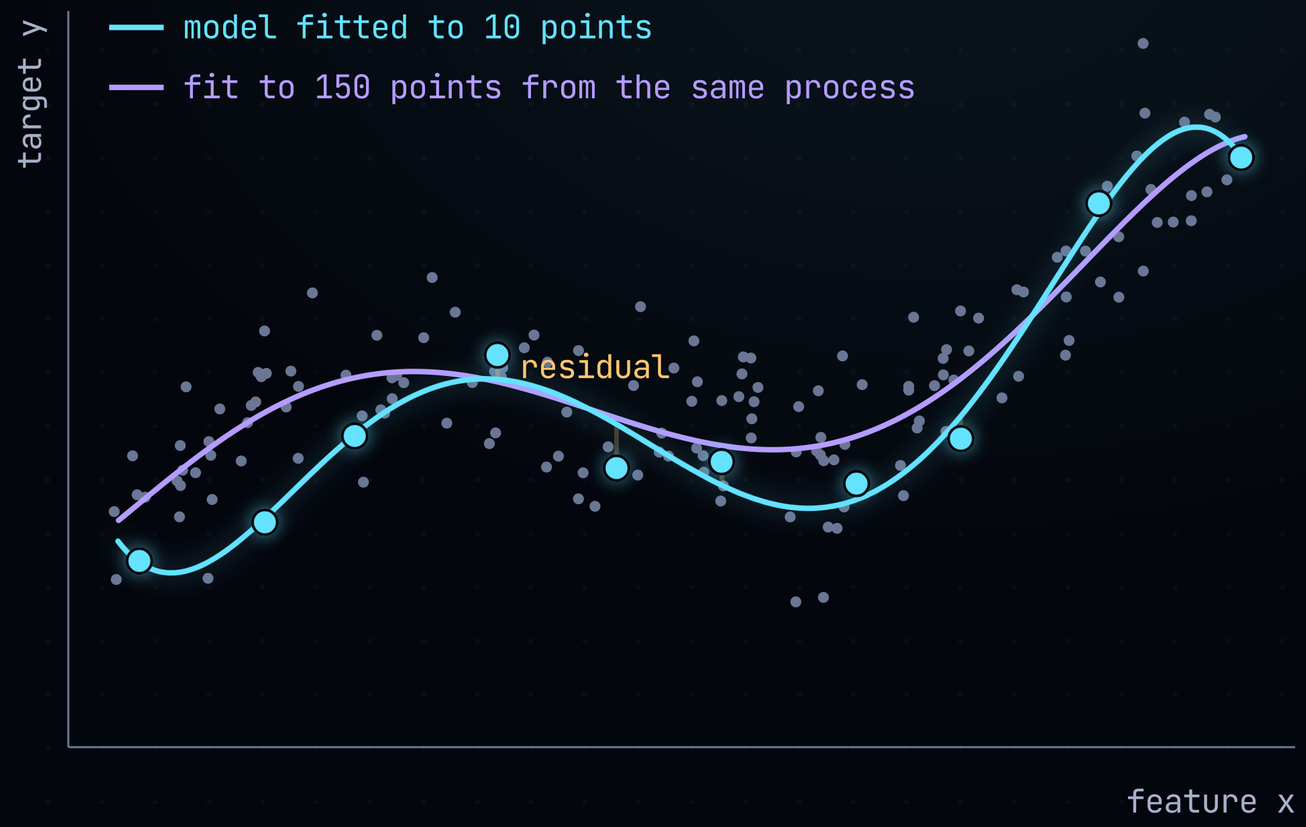
REGRESSION

Predict a *number* from a row of features — e.g. a building's heating load from its geometry.

RESIDUAL

What the model got wrong for that row: observed − predicted.

$$z_i = y_i - \hat{g}(x_i)$$



Illustrative example — synthetic points, real least-squares fits

Problem framing: Mohebbi et al. (2026), §1 · Plot: illustrative synthetic data generated by us

TODO: Member A

02 — WHY NOT JUST AUGMENT?

Images have safe transformations. Tables don't.

Images: a label-preserving symmetry



LABEL "7" ✓ still true after rotating

Tabular regression: no obvious symmetry

FLOOR AREA	BEDS	GARAGE	LOCATION	PRICE (TARGET)
92 m²	3	basic	zone B	£285,000
110 m²	3	basic	zone B	£285,000

NAIVE COPY floor area +20%, price copied over

✗ Bigger house, same price — the label no longer fits the row

Illustrative example — invented values

INTERPOLATE ROWS

SMOTE-style, C-Mixup (Chawla et al., 2002; Yao et al., 2022)

GENERATE ROWS

CTGAN, TabDDPM — target treated as just another column (Xu et al., 2019; Kotelnikov et al., 2023)

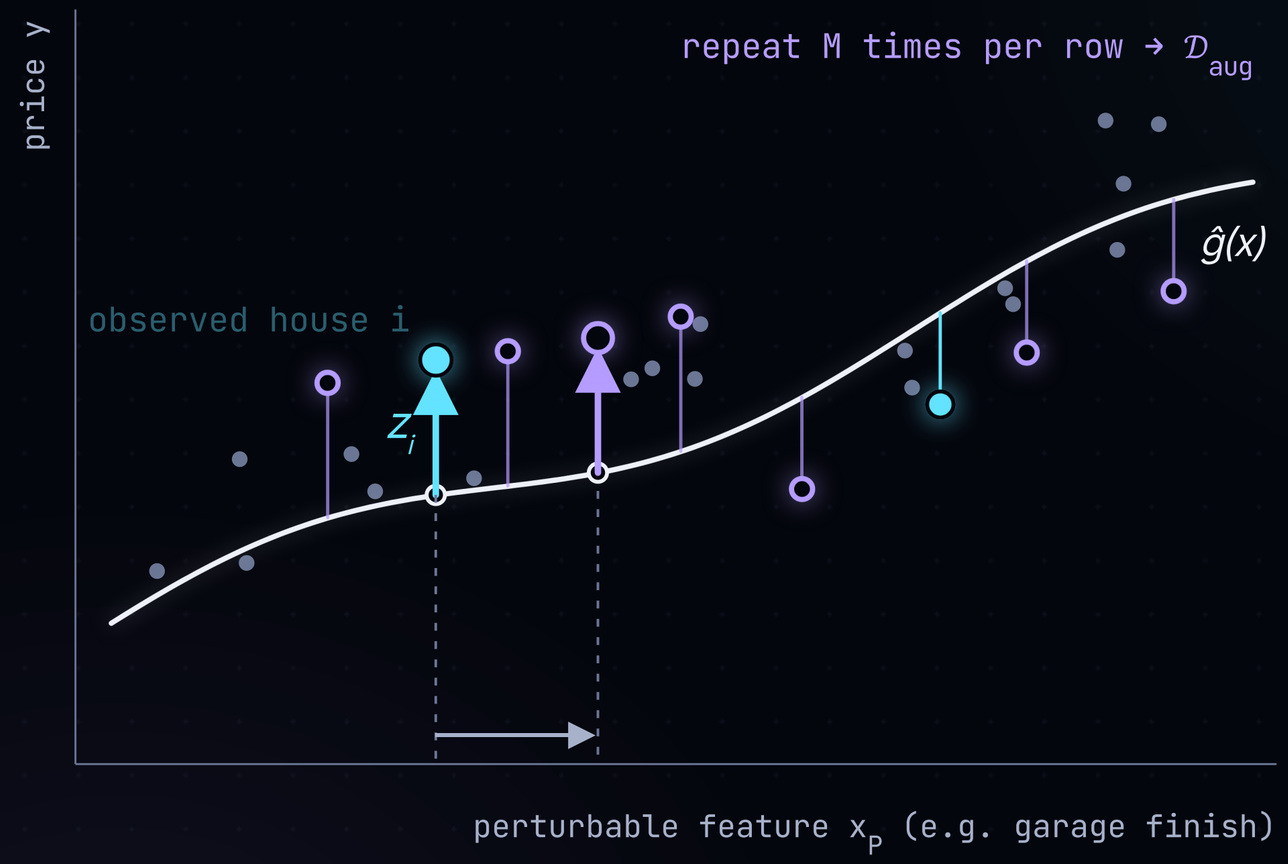
AUTHORS' ARGUMENT

neither keeps each row's *own* residual noise

Row: illustrative, invented values · Approaches as characterised by Mohebbi et al. (2026), §2

TODO: Member A

Move the safe features. Keep the observed noise.



Illustrative 1-D sketch — the method perturbs up to *k* screened features of a multi-feature row

1 Fit a baseline

$\hat{g} \leftarrow$ MLP or XGBoost on $\mathcal{D}_{\text{train}}$

2 Compute each residual

$z_i = y_i - \hat{g}(x_i)$

3 Perturb selected features only

$x'_{i,P} = x_{i,P}\,(1+\delta), \quad \delta \sim \mathcal{U}[-p,\, p]$

$x_{i,R}$ held fixed

4 Reuse the same residual

$y'_i = \hat{g}(x'_i) + z_i$

X_R fixed: location, floor area

X_P perturbed: garage finish

Mohebbi et al. (2026), Alg. 1 & §4.1; house example §1 · Plot: illustrative 1-D sketch

TODO: Member B

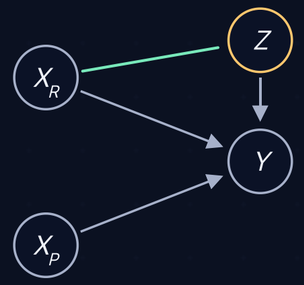
04 — WHAT MUST BE TRUE

It only works if the noise ignores the features we move

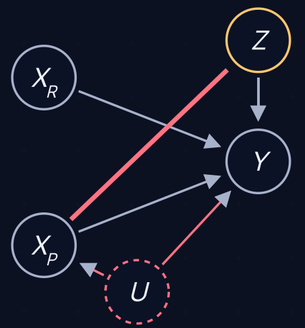
$$P(Z \mid X_P, X_R) = P(Z \mid X_R)$$

Given the fixed features, the residual carries no information about the perturbed ones.

`Assumption 3.1 — assumed and screened for, not guaranteed`



✓ holds X_P moves Y, not Z. Z may depend on X_R.



✗ breaks hidden U drives X_P and Y; Z absorbs U.

F1 ~~F2~~ F3 F4 ~~F5~~ F6 ~~F7~~ F8 illustrative features

PC CAUSAL-GRAPH CHECK
drop features with an edge to $\hat{z}$

PEARSON SCREEN
drop features correlated with $\hat{z}$ (repo default $|r| \geq 0.1$)

survivors → X_P

WILCOXON SAFETY GATE
10-fold CV, augmented vs. unaugmented errors
$p < 0.05$ → use CRDA otherwise → keep baseline

Also fails if ĝ is inaccurate: its errors leak into ẑ and stay tied to X_P (§3.3).

Screens are heuristics: PC can miss hidden confounders; Pearson misses non-linear links (§6).

Mohebbi et al. (2026), Assump. 3.1, Fig. 1 adapted, §3.3, §6 · Spirtes et al. (2000) · Wilcoxon (1945)

TODO: Member B

A focused, checkable reproduction — not a re-invention

1. **Install & audit**
   authors' repo, Python 3.11, pinned `requirements.txt`
2. **Run the demo**
   `demo/demonstration.ipynb` — Energy Efficiency
3. **Narrow the scope**
   Energy Efficiency + Concrete Strength · MLP & XGBoost
4. **Controlled comparison**
   unaugmented vs CRDA on identical splits & seeds (TODO: e.g. 3 seeds first pass (paper: 15))
5. **Compare with published**
   our ΔMSE next to Table 1, matching their protocol
6. **Document**
   differences, obstacles, versions, runtimes

PROPOSED TARGETS — NOT COMPLETED WORK · Proposed — not yet started

Reported by the original paper — **not yet replicated by our group**

−22.9% MLP **−6.4%** XGBoost

average test-MSE change across 9 benchmark datasets



Our comparison targets: Table 1, mean of 15 seeds, largest subset size

QUESTIONS FROM READING THE CODE — TO VERIFY, NOT CONCLUSIONS

? Table 1 configs set `ignore_filter: true`, so ΔMSE is measured with augmentation forced on — the Wilcoxon decision is only logged
? Code scores the augmented model as the average of 10 CV-fold models; Alg. 1 describes one retrained model

Clear ownership, balanced load, every result cross-checked

A — PLANNED

TODO: Member A

Environment & reproducibility

- Python 3.11 env from the authors' pinned requirements
- Run the demo notebook end-to-end
- Seed, config and version logging

CROSS-CHECK

Re-runs one of B's configs independently

SPEAKS

slides 1, 2, 3 + own card

B — PLANNED

TODO: Member B

Experiment execution

- Baseline vs CRDA runs (MLP, XGBoost)
- Controlled splits and seeds
- Saves every config.json / results.csv

CROSS-CHECK

Recomputes C's comparison table from raw results

SPEAKS

slides 4, 5 + own card

C — PLANNED

TODO: Member C

Analysis, documentation & delivery

- Our ΔMSE vs the paper's Table 1
- Obstacle and deviation log
- Slides, timing and rehearsal

CROSS-CHECK

Rebuilds A's environment from the written steps

SPEAKS

slides 6, 8, 9 + own card



Shared by everyone: run at least one config end-to-end · review the final slides · rehearse to time

07 — TARGETS BEFORE WEEK 9

What we will bring to the next update

Week 5 · today — Week 9 · progress update

1 — Environment reproduced

TODO: Week 6

DONE WHEN

Authors' demo notebook runs end-to-end on our machine

EVIDENCE

pip freeze, OS / CPU note, demo output

TODO: Member A

2 — Baselines logged

TODO: Week 7

DONE WHEN

Unaugmented MLP + XGBoost on dataset 1, fixed seeds

EVIDENCE

config.json + results.csv per run

TODO: Member B

3 — First CRDA comparison

TODO: Week 8

DONE WHEN

Our ΔMSE next to the paper's Table 1 row

EVIDENCE

Side-by-side table, Wilcoxon p-values

TODO: Member B TODO: Member C

4 — Obstacle log

Ongoing

DONE WHEN

Every deviation recorded with its fix or open question

EVIDENCE

Shared log, discussed at Week 9

TODO: Member C

OUR RESEARCH QUESTION

Do CRDA's reported gains hold when we re-run the authors' own pipeline — and where do they not?

Targets and dates: editable in src/config/deck.config.ts · all milestones currently planned

REFERENCES

Sources

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All URLs checked 19 October 2026

TODO: Member C